

Question

The following compound **A** can alleviate the repulsion between lone pair electrons of nitrogen atoms when binding to a proton, and simultaneously form intramolecular hydrogen bonds. Therefore, **A** has a strong basicity (the pK_a of its conjugate acid is 12.1), and is called a "proton sponge":

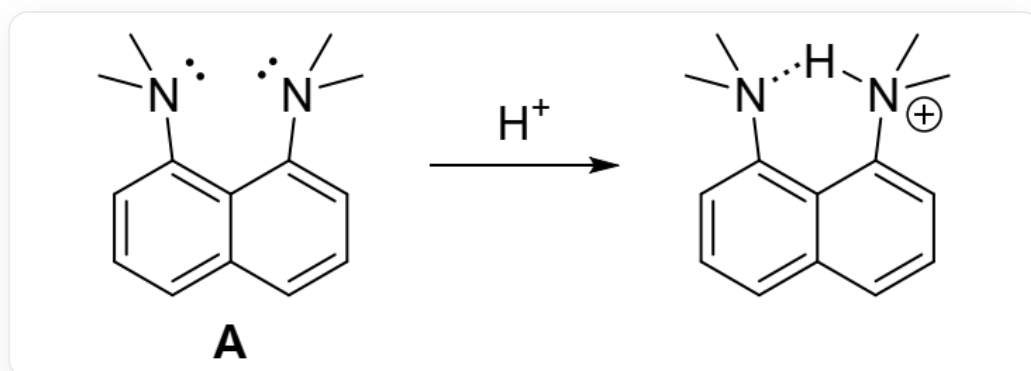
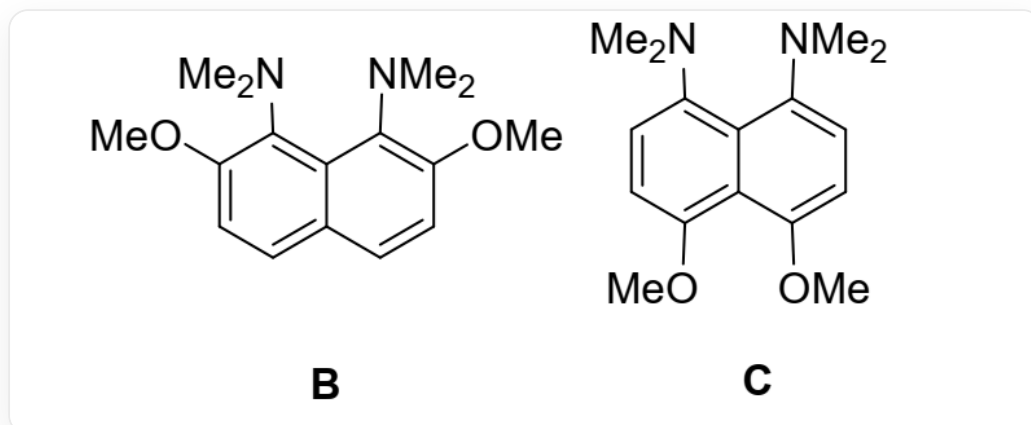


Fig 1: The SMILES code of this reaction is: CN(C)C1=CC=CC2=C1C(N(C)C)=CC=C2>[H+]>CN(C)C3=CC=CC4=C3C([N+](C)([H])C)=CC=C4

Further studies have found that the introduction of methoxy groups at the ortho position of dimethylamino can greatly enhance its basicity. For example, the pK_a of the conjugate acid of **B** in the following compounds is 16.1, while the pK_a of the conjugate acid of **C** is 13.9:



The SMILES code of B is: CN(C)C1=C(OC)C=CC2=C1C(N(C)C)=C(OC)C=C2, C is : CN(C)C1=CC=C(OC)C2=C1C(N(C)C)=CC=C2OC

Which of the following statements are correct:

1. The nitrogen atom in **B** is close to sp^2 hybridization.
2. The methoxy group has an electron-donating inductive effect, which increases the electron cloud density on the nitrogen atom, making the nitrogen atom easier to bind to a proton.
3. In the comparison of the basicity of **A** and **B**, the electronic effect is a more important factor affecting the basicity of **B** than the steric effect.
4. The difference in basicity between **B** and **C** mainly comes from the steric hindrance effect provided by the methoxy group. The methoxy group and dimethylamino group are relatively crowded, resulting in the two dimethylamino groups of **B** being closer, the repulsion of lone pair electrons on the nitrogen atom being greater, and the combination with a proton can alleviate more repulsion.

A. All other options are incorrect

B. 1.

C. 2.

D. 3.

E. 4.

F. 1 & 2.

G. 1 & 3.

H. 1 & 4.

I. 2'3.

J. 2'4.

K. 3'4.

L. 1'2'3.

M. 1'2'4.

N. 1'3'4.

O. 2'3'4.

P. 1'2'3'4.

Answer

Correct Answer: **H**

Detailed Explanation

The conjugate acid of compound **A** has a $pK_a = 12.1$, while the conjugate acid of **C** has a $pK_a = 13.9$.

The difference between the two arises from the para-methoxy group to the dimethylamino group. Since the para-methoxy group cannot produce steric hindrance to the dimethylamino group, the difference in pK_a of its conjugate acid is entirely due to the electron-donating resonance effect of the methoxy group, which is approximately 1.8 orders of magnitude in the pK_a of the conjugate acid.

CHECKPOINT

1 PTS

The difference in basicity between **A** and **C** arises from the electron-donating resonance effect of the methoxy group.

For a conjugated system, adding one or a few double bonds does not significantly weaken the resonance effect of the substituent. Therefore, the electron-donating resonance effect of the methoxy group in **B** and **C** is somewhat reduced when it is in the para position compared to the ortho position, but the impact is not significant.

The conjugate acid of compound **B** has a $pK_a = 16.1$, while the conjugate acid of **C** has a $pK_a = 13.9$. Therefore, the difference in basicity between **B** and **C** can be largely attributed to the steric hindrance effect brought about by the ortho-methoxy group.

CHECKPOINT

1 PTS

The position of the methoxy group, whether ortho or para, has little effect on the electron-donating resonance effect.

CHECKPOINT

1 PTS

The difference in basicity between **B** and **C** can be largely attributed to the steric hindrance effect brought about by the ortho-methoxy group.

The methoxy group and the dimethylamino group are relatively crowded, resulting in the two dimethylamino groups of **B** being closer to each other, and the repulsion of lone pair electrons on the nitrogen atom being greater, which can alleviate more repulsion when binding to a proton.

CHECKPOINT

1 PTS

Binding a proton can alleviate more repulsion, 4 is correct.

At the same time, due to this repulsive effect, the two methyl groups of the dimethylamino group, when the nitrogen atom adopts sp^3 hybridization, will be repelled by the ortho substituent or another dimethylamino group regardless of the conformation. Therefore, in this case, it will tend to sp^2 hybridization to some extent.

CHECKPOINT

1 PTS

When the nitrogen atom adopts sp^3 hybridization, the two methyl groups of the dimethylamino group will be repelled by the ortho substituent or another dimethylamino group. 1 is correct.

Methoxy group has an electron-donating resonance effect. Among them, oxygen has an electron-withdrawing inductive effect due to its large electronegativity, but the overall electron-donating resonance is greater than the electron-withdrawing inductive effect and dominates.

CHECKPOINT

1 PTS

The electron-donating resonance effect of the methoxy group is greater than the electron-withdrawing inductive effect, 2 is incorrect.

Returning to the comparison of **A** and **B**, it is not difficult to know that the magnitude of the effect of the electron-donating resonance effect on the conjugate acid pK_a is slightly greater than 1.8, and the rest are the effects of steric hindrance. Therefore, the steric hindrance effect is greater than the electronic effect.

CHECKPOINT

1 PTS

The steric hindrance effect is greater than the electronic effect, 3 is incorrect.

In summary, the answers are 1, 4.